

Department of Data Science

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Project Seminar on

BRAIN TUMOR SEGMENTATION IN MRI USING ATTENTION-BASED CNNs

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BRAIN TUMOR SEGMENTATION IN MRI USING ATTENTION-BASED CNNs

INTRODUCTION

- Gliomas and other brain tumors are among the most aggressive and heterogeneous cancers.
- Accurate delineation of tumor sub-regions from MRI is central to diagnosis, treatment planning, and monitoring.
- Manual segmentation by radiologists:
 - Time-consuming — takes 30–60 minutes per patient scan
 - Requires specialised expertise and suffers from inter-observer variability
 - Limits scalability in high patient-volume clinical settings
- CNNs (U-Net style encoder-decoders) are the de facto standard, but plain convolutions treat every location/channel equally — struggling with irregular shapes and blurred tumor boundaries.
- This project designs a 3D attention-augmented CNN pipeline for multimodal MRI tumor segmentation, trained on the multi-institutional BraTS dataset using multi-GPU DGX infrastructure.

AIMS AND OBJECTIVES

- **Aim:**

- Design, train, and evaluate an attention-based 3D CNN for multi-class brain tumor sub-region segmentation from multimodal MRI.

- **Objectives:**

- Study and compare existing CNN and attention-based architectures for volumetric segmentation of images
- Design a 3D attention-augmented encoder-decoder with channel + spatial attention gates on skip connections
- Implement a complete pre-processing and patch-based training pipeline for BraTS
- Train the model using multi-GPU distributed training on the college's DGX system
- Evaluate against U-Net and Attention U-Net baselines using Dice, IoU, Hausdorff95, sensitivity, specificity
- Visualize and interpret predictions (attention maps, Grad-CAM) for clinical plausibility

ABOUT THE DATASET — BRaTS

- BraTS — Brain Tumor Segmentation Challenge, run since 2012 with MICCAI, RSNA and ASNR.
- Multi-institutional: MRI scans from 80+ hospitals worldwide, reflecting real-world scanner/protocol variability.
- ~1,251 training cases, each a 3D volume of $240 \times 240 \times 155$ voxels at 1mm^3 isotropic resolution.
- 4 MRI modalities per patient: T1 (anatomy), T1ce (active tumor), T2 (fluid/swelling), FLAIR (edema).
- Ground truth: necrotic core, peritumoral edema, enhancing tumor — combined into Whole Tumor (WT), Tumor Core (TC), Enhancing Tumor (ET).
- Challenges: severe class imbalance, heterogeneous scanner quality, irregular/blurry tumor boundaries.
- Preprocessing: skull-stripping, N4 bias correction, intensity normalisation, co-registration, resampling, overlapping 3D patching.

SYSTEM / PIPELINE ARCHITECTURE

- 1 Multimodal MRI Input (T1, T1ce, T2, FLAIR)
- 2 Preprocessing (skull-stripping, N4 correction, normalisation, resampling)
- 3 3D Attention-Augmented Encoder-Decoder CNN (channel + spatial attention on skip connections)
- 4 GPU Training (Dice + cross-entropy loss, patch sampling, mixed precision)
- 5 Post-processing (connected-component filtering, morphological clean-up)
- 6 Output: Segmented Tumor Sub-regions (WT, TC, ET)
- 7 Evaluation vs Ground Truth (Dice, IoU, HD95, Sensitivity, Specificity)

PROPOSED METHODOLOGY

- Pre-processing of multimodal MRI volumes: skull-stripping, N4 bias-field correction, intensity normalisation, co-registration, isotropic resampling.
- Volumes divided into overlapping 3D patches to fit GPU memory while preserving volumetric context.
- Core architecture: 3D encoder-decoder CNN where every skip connection passes through a channel attention block (squeeze-and-excitation) followed by a spatial attention gate.
- Trained end-to-end with combined Dice + weighted cross-entropy loss to address class imbalance.
- Multi-GPU data-parallel training with mixed precision on the college DGX system.
- Model selection via k-fold cross-validation on the BraTS training split, evaluated on the held-out validation set.
- Ablation study isolates channel attention, spatial attention, and the combined hybrid block against a plain 3D U-Net baseline.

TECHNOLOGY REQUIREMENTS

- **Hardware:**

- RCOEM DGX system — NVIDIA B200 (Blackwell), 23 GB VRAM, multi-GPU distributed training

- **Software / Libraries:**

- Language: Python 3, PyTorch / MONAI (3D medical image deep learning)
- NumPy, NiBabel, SimpleITK (MRI pre-processing)
- scikit-learn (evaluation metrics), Matplotlib / 3D Slicer (visualisation)
- CUDA-based multi-GPU distributed training

APPLICATIONS & ADVANTAGES

Applications

- Pre-screening / triage support for radiologists
- Second-opinion overlay in tumor board reviews
- Longitudinal treatment-response monitoring
- Research tool for computational neuro-oncology
- Template pipeline for other volumetric segmentation tasks

Advantages

- Attention focuses computation on tumor-relevant regions
- Lightweight — no transformer bottleneck, lower compute cost
- Improved boundary precision for small enhancing-tumor regions
- Interpretable via attention-map and Grad-CAM visualisation
- Ablation-validated — each component's contribution is proven, not assumed

EXPECTED OUTCOMES & DELIVERABLES

- Trained 3D attention-augmented CNN for multi-class tumor sub-region segmentation.
- Quantitative benchmarking (Dice, IoU, HD95, sensitivity, specificity) against U-Net and Attention U-Net baselines.
- Complete pre-processing and training codebase, plus trained model checkpoints.
- Ablation study report isolating channel / spatial / combined attention contributions.
- Attention-map and Grad-CAM visualisations for interpretability.
- Lightweight demo interface overlaying predicted segmentation masks on MRI slices.
- Target: Dice ~ 0.87 on Whole Tumor — competitive with published Attention U-Net work, without transformer-scale compute.

PROJECT TIMELINE

Week	Phase	Milestone
1–3	Literature Review	Dataset access, literature survey, problem formulation
4–6	Preprocessing & Baseline	Pre-processed volumes + working baseline 3D U-Net
7–10	Attention Module Design	Channel + spatial attention integrated into encoder-decoder
11–13	DGX Training ← We are here	Trained checkpoints, logged Dice / HD95 across folds
14–15	Evaluation & Ablation	Quantitative comparison vs U-Net / Attention U-Net baselines
16	Documentation	Final report, demo interface, synopsis submission

CONCLUSION

- Attention mechanisms let CNNs focus computation on tumor-relevant regions instead of treating the whole brain volume equally.
- Integrating channel and spatial attention into every skip connection improves boundary precision without transformer-scale compute.
- The ablation study will provide rigorous, scientific proof of each attention component's individual contribution.
- Training on the RCOEM DGX (NVIDIA B200) makes full 3D volumetric training feasible within the semester timeline.
- The proposed pipeline lays the foundation for a lightweight, interpretable, and clinically-plausible segmentation framework.

THANK YOU